MIL-F-28675B
10 June 1985
SUPERSEDING
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16 November 1979

MILITARY SPECIFICATION

FITTINGS, CONDUIT, EXPLOSION PROOF (COUPLINGS, SEALS, UNIONS, ELBOWS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers explosion-proof electrical conduit fittings including conduit seals, flexible couplings, reducers, unions, drain and breather valves, inspection seals, drain seals, and elbows.

1.2 Classification.

1.2.1 Types, classes, and styles. The fittings shall be one of the following types, classes, and styles as specified (see 6.2.1):

Type I - Conduit seals.

Class 1 - Vertical.
Class 2 - Vertical/horizontal.
Class 3 - Multi-angular.

Style A - Female hub, top and bottom.
Style B - Male hub, top and female, bottom.
Style C - Female hub, top and male, bottom.
Style D - Elbow, female hub, top and bottom.

Type II - Flexible couplings.

Class 1 - Female with removable male nipple and female union.
Class 2 - Female with removable male nipples, both ends.
Class 3 - Male threaded end fittings.
Class 4 - Male threaded end fitting on one end. Female union on the other.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5975

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Type III - Reducer.
Type IV - Unions.

Class 5 - Male.
Class 6 - Female.
Class 7 - Angle, 90-degree (°) elbow.
Class 8 - Universal, male.
Class 9 - Universal, female.

Type V - Drain and breather valve.
Type VI - Inspection seal.
Type VII - Drain seal.
Type VIII - Elbows.

Class 10 - 90° male.
Class 11 - 90° female.
Class 12 - 45° female.
Class 13 - 90° male and female.
Class 14 - Flexible.

1.2.2 Size. Fitting sizes shall be 1/8, 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, or 6 inch sizes as specified (see 6.2.1). Fitting sizes as related to type, class, and style shall be as shown in tables I through VII.

TABLE I. Type I conduit seals.

Style	Class 1 sizes (inches)	Class 2 sizes (inches)	Class 3 sizes (inches)
A	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3
B	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3,
C	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3,
D	3/4	3/4	

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TABLE II. Type II flexible couplings.

Flexible length (inches)	All classes Size (inches)
4	1/2, 3/4
6	1/2, 3/4, 1
8	1/2, 3/4, 1
10	1/2, 3/4, 1
12	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
15	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
18	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
21	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
24	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
27	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
30	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
33	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4
36	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4

TABLE III. Type III reducers.

Male thread Size (inches)	Female thread Size (inches)	Male thread Size (inches)	Female thread Size (inches)
1/2	1/8	2-1/2	1
1/2	1/4	2-1/2	1-1/4
1/2	3/8	2-1/2	1-1/2
3/4	1/2	2-1/2	2
1	1/2	3	1
1	3/4	3	1-1/4
1-1/4	1/2	3	1-1/2
1-1/4	3/4	3	2
1-1/4	1	3	2-1/2
1-1/2	1/2	3-1/2	2
1-1/2	3/4	3-1/2	2-1/2
1-1/2	1	3-1/2	3
1-1/2	1-1/4	4	2
2	1/2	4	2-1/2
2	3/4	4	3
2	1	4	3-1/2
2	1-1/4	5	4
2	1-1/2	6	5

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TABLE IV. Type IV unions.

Class	Size (inches)
5	1/2, 1/2 to 3/4, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6
6	1/2, 3/4 to 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6
7	1/2 to 1/2, 1/2 to 3/4, 3/4 to 1/2, 3/4 to 3/4
8	1/2, 3/4, 1
9	1/2, 3/4, 1

TABLE V. Type V drain and breather valves.

Size (inches)
1/4
3/8
1/2 (straight)
1/2 (90° elbow)

TABLE VI. Type VI inspection seals and type VII drain seals.

Type	Hub size (inches)
VI	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
VII	1/2, 3/4, 1/ 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4

TABLE VII. Type VIII elbows.

Class	Size (inches)
10	1/2, 3/4, 1, 1-1/4
11	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2
12	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4
13	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
14	1/2, 3/4, 1

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1.2.3 Design installation use. When required for a specific application in an existing or planned installation, the manner in which the fittings will be used shall be specified (see 6.2.1).

Use (a) - Vertical only.
Use (b) - Horizontal only.
Use (c) - Horizontal or vertical.
Use (d) - Any angle between horizontal and vertical.

1.2.4 Hazardous locations. Fittings shall be specified for intended use by locational class and group designations (see 3.1 and 6.2.1).

1.2.5 Part number. The military specification part number for items described in this specification shall be identified as shown in 6.6.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 886 - Outlet Boxes and Fittings for Use in Hazardous Locations, Class I, Groups A, B, C, and D, and Class II, Groups E, F, and G.
Hazardous Location Equipment Directory.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification establishes the construction and acceptance requirements for explosion-proof conduit fittings. These fittings shall comply with UL 886 for use in Class I, Division I, Groups A, B, C, and D and Class II, Division I, Groups E and G hazardous locations. The class, division, and group definitions are as defined in NFPA 70.

3.2 Standard commercial product. The conduit fittings shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the fittings being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2.1), the contractor shall furnish one fitting of each classification included in the contract for first article inspection and approval (see 4.2.1 and 6.3).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using

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materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified. The fittings shall be made of materials as specified in UL 886.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts and assemblies.

3.6 Construction. Conduit fittings shall be constructed in a manner to ensure safe use in the hazardous location for which the fittings are intended to be used. Minimum wall thickness of the fittings shall be as specified in UL 886. Conduit threads shall conform to FED-STD-H28 and UL 886.

3.7 Codes and standards. The conduit fitting shall conform to the requirements of UL 886.

3.7.1 Compliance. Prior to approval of the first shipment, the contractor shall submit for the approval of the contracting officer, or his authorized representative, satisfactory evidence that the fittings he proposes to furnish under this specification meet the requirements of UL 886.

3.7.1.1 UL 886. Acceptable evidence of meeting the requirements of UL 886 shall be the UL certification symbol or label, listing in the UL Hazardous Location Equipment Directory, or a certified test report (see 6.2.2) from a recognized independent testing laboratory indicating the fittings have been tested and conform to UL 886. Such evidence must be acceptable to the contracting officer.

3.8 Identification markings. Unless otherwise specified (see 6.2.1), identification markings shall be permanently and legibly marked on each fitting. The markings shall include, but not limited to, the applicable information specified in NFPA 70 for hazardous locations.

3.9 Workmanship. The fittings shall be of the highest quality, free from any defects that would prevent the fitting from conforming to the requirements of this specification, or that would degrade the appearance of the fitting, such as dirt, grease, or grime. Threaded portions of the fitting shall be clean and undamaged.

3.10 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one fitting of each classification included in the contract when a first article is required (see 3.3 and 6.3). This inspection shall include the examination of 4.4, the tests of 4.5 and, when specified, the first article pack of 4.6 (see 4.6 and 6.2.1). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. When specified (see 6.2.1), sampling and inspection procedures shall be in accordance with MIL-STD-105. All fittings of the same classification offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level shall be 2.5 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots. The contracting officer may waive sample requirements when certification as specified in 3.7.1.1 is furnished.

4.4 Examination. Each fitting selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for

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adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.4.1 Standard compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard cited in 3.7. The Government reserves the right to examine and test all fittings to determine the validity of the certification.

4.5 Tests. The fittings shall be tested in accordance with UL 886. The acquiring activity shall specify the specific tests to be performed (see 6.2.1). Failure to pass any test shall be cause for rejection.

4.6 Packaging inspection. The inspection of the preservation, packing, palletization, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a first article pack shall be furnished for examination and test within the timeframe required (see 6.2.1).

5. PACKAGING

5.1 Preservation, packing, palletization, and marking. Preservation, packing, palletization, and marking shall be in accordance with the requirements of MIL-E-17555 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The electrical conduit fittings covered by this specification are intended for use as follows:

Type I. To prevent the passage of gases of combustion and flames from one portion of the electrical installation to another through conduit.

Type II. To connect equipment where difficult bends are required or where the equipment is subject to vibration or movement.

Type III. To reduce a conduit hub to a smaller size.

Type IV. To connect conduit to a conduit fitting, junction box, or device enclosure.

Type V. To allow accumulated water to drain out of an enclosure and to provide ventilation to minimize condensation in an enclosure.

Type VI. To provide a conduit seal with a removable cover for inspection.

Type VII. To provide a conduit seal with a removable inspection cover and a drain to prevent accumulation of water above the seal.

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Type VIII. To connect conduit to a conduit fitting, junction box, or device enclosure when a turn is required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style required (see 1.2.1).
c. Size required (see 1.2.2).
d. Required use (when applicable) (see 1.2.3).
e. Explosion-proof criteria required in accordance with hazardous location classes and groups (see 1.2.4).
f. Issue of document required if other than as specified (see 2.1.1).
g. When a first article is required for inspection and approval (see 3.3, 4.2.1, and 6.3).
h. When identification markings shall be other than as specified (see 3.8).
i. When a first article pack inspection is required and the timeframe required for submission (see 4.2.1 and 4.6).
j. When sampling in accordance with MIL-STD-105 is required (see 4.3).
k. Specified tests of UL 886 required for first article test (see 4.5).
l. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.
3.7.1.1	Reports, Test	DI-T-2072
4.4.1	Certificate of Compliance	DI-E-2121

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one fitting of each

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classification. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Special requirements. It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If for any particular purpose samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid samples should be definitely stated. The specification shall apply in all other respects. Fittings required for replacements in existing systems wherein dimensional tolerances are critical, shall be specified as to manufacturer (or otherwise detailed) to ensure ease of replacement actions at minimum cost.

6.5 Specification applicability. Military specifications do not necessarily include all types, designs, forms, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.6 Definitive Military Specification part number. The Military Specification part number is a definitive part number which corresponds to the type, class, and size of fittings covered by this specification, and defines the requirements of the options presented under this specification.

6.6.1 Cataloging data. For cataloging purposes, part numbers for the fittings are assigned as follows:

	M28675	X	X	X	X
Military specification number					
Size and type code number (see 6.6.2)					
Class code letter (see 6.6.3)					
Hazardous location code letter (see 6.6.4)					
Design installation use number (see 6.6.5)					

6.6.2 Size and type. The type of fitting and size (see 1.2.1 and 1.2.2) are identified by a number and letter (see table VIII).

Table VIII. Code number and letter to size and type.

Fitting size (nominal)	Type of fittings							
	I	II	III	IV	V	VI	VII	VIII
1/8	1A	2A	3A	4A	5A	6A	7A	8A
1/4	1B	2B	3B	4B	5B	6B	7B	8B
3/8	1C	2C	3C	4C	5C	6C	7C	8C
1/2	1D	2D	3D	4D	5D	6D	7D	8D
3/4	1E	2E	3E	4E	5E	6E	7E	8E
1	1F	2F	3F	4F	5F	6F	7F	8F

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Table VIII. Code number and letter to size and type. (Continued)

Fitting size (nominal)	Type of fittings							
	I	II	III	IV	V	VI	VII	VIII
1-1/4	1G	2G	3G	4G	5G	6G	7G	8G
1-1/2	1H	2H	3H	4H	5H	6H	7H	8H
2	1I	2I	3I	4I	5I	6I	7I	8I
2-1/2	1J	2J	3J	4J	5J	6J	7J	8J
3	1K	2K	3K	4K	5K	6K	7K	8K
3-1/2	1L	2L	3L	4L	5L	6L	7L	8L
4	1M	2M	3M	4M	5M	6M	7M	8M
5	1N	2N	3N	4N	5N	6N	7N	8N
6	1O	2O	3O	4O	5O	6O	7O	8O

6.6.3 Class. The fitting class (see 1.2.1) is identified by a single letter (see table IX).

TABLE IX. Code letter to class of fitting.

Code letter	Fitting class
A	Class 1
B	Class 2
C	Class 3
D	Class 4
E	Class 5
F	Class 6
G	Class 7
H	Class 8
I	Class 9
J	Class 10
K	Class 11
L	Class 12
M	Class 13
N	Class 14

6.6.4 Hazardous location. Class and group of hazardous location (see 3.1) shall be identified by a single letter (see table X).

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TABLE X. Code letter to hazardous location.

Code letter	Class and group of hazardous location
A	Class I Group A
B	Class I Group B
C	Class I Group C
D	Class I Group D
E	Class II Group E
G	Class II Group G
H	Class III

6.6.5 Design installation use. The use of the fitting (see 1.2.3) shall be identified by a single digit number (see table XI).

TABLE XI. Code number to installation use.

Code number	Installation use
1	Vertical only
2	Horizontal only
3	Horizontal or vertical
4	Any angle between horizontal and vertical

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ER
Navy - YD
Air Force - 85

Preparing Activity:

Navy - YD

(Project 5975-0605)

Review Activities:

Army - ME
Air Force - 99
DLA - GS

User Activity:

Army - AR

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